

Dr. Fabian Kohlmann

Chief Executive Officer



GDAC-SA Tender Role: Managing Director | Global PaleoDEM Development Expert | 15+ Years Halliburton | Saudi Arabia Experience | Neflex Product Lead

Chief Executive Officer

GDAC-SA Tender Role

Managing Director | Global PaleoDEM Development Expert | 15+ Years Halliburton | Saudi Arabia Experience | Neflex Product Lead

Current Position

Managing Director

Lithodat Pty Ltd | Melbourne, Australia 2024 - Present

- Strategic leadership and overall company direction
- Driving company vision and growth strategy
- Leading GDAC-SA tender response and international business development
- Industry-recognized leader in data geosciences and exploration data solutions

Director

Lithodat Pty Ltd | Melbourne, Australia 2019 - 2024 (5 years)

- Co-founded Lithodat with focus on geoscience data management
- Built company from startup to established consultancy
- Developed core product strategy and client relationships
- Technical leadership for all major projects

Core Expertise

- Petroleum Geodynamics & Basin Analysis
- Petroleum Systems Modeling
- Structural Geology & Tectonics
- Geoscience Software Development
- Team Leadership & Mentoring
- Client Relations & Business Development

Key Professional Experience

Product Lead - Neflex Geodynamics

Halliburton | United Kingdom 2015 - 2019 (4 years)

- Led global product development for Neflex geodynamics suite
- Managed international team of geoscientists and developers
- Defined product roadmap based on client needs and market analysis
- Direct client engagement for major oil & gas companies globally
- Responsibility for product P&L and strategic direction

Senior Geodynamics Specialist

Halliburton (Neflex) | United Kingdom 2010 - 2015 (5 years)

- Advanced petroleum systems modeling and basin analysis
- Client-facing technical consulting for major exploration projects
- Developed innovative workflows for basin evaluation
- Training and mentoring of junior geoscientists
- Contributed to software development and QA processes

Geodynamics Specialist

Halliburton (Neftek) | United Kingdom 2005 - 2010 (5 years)

- Petroleum systems modeling for diverse basins globally
- Structural restoration and kinematic modeling
- Integration of geophysical and geological data
- Client project delivery and technical support
- Development of best practices and methodologies

Education

PhD - Structural Geology

University of Adelaide | Adelaide, Australia 2001 - 2005

- Thesis focused on structural controls on petroleum systems
- Advanced research in basin analysis and tectonics
- Published research in peer-reviewed journals

Master of Science (MSc) - Geology/Earth Science

Ludwig-Maximilians-Universität München | Munich, Germany 1999 - 2003

- General Geology/Earth Science

BSc (Hons) - Geology

University of Adelaide | Adelaide, Australia 1997 - 2000

- First Class Honours in Geology
- Strong foundation in structural geology and tectonics

Technical Skills

- **Geodynamics Software:** Expert in basin modeling, structural restoration, petroleum systems analysis
- **GIS & Mapping:** Advanced GIS skills, geological mapping, spatial analysis
- **Programming:** Experience with Python, SQL, geoscience software development
- **Data Management:** Geoscience data workflows, quality control, database design
- **Project Management:** Team leadership, client engagement, project delivery

Geographic Expertise

- **Primary:** Australia, Southeast Asia, Middle East (including Saudi Arabia)
- **Secondary:** North Sea, North Africa, South America
- **Global Experience:** Worked on projects across all major petroleum basins

Relevance to GDAC-SA Tender

- **Strategic Leadership:** Managing Director with proven track record building Lithodat from startup to established consultancy
- **Halliburton Connection:** 15+ years at Halliburton (NefTex) - same company background as Dr. Qusay Abeed, ensuring cultural fit and shared methodology
- **Product Leadership:** Led global geodynamics product line - brings product management expertise critical for platform development
- **Saudi Arabia Experience:** Direct experience working with Saudi clients and understanding of regional geology
- **Global Perspective:** International experience across UK, Norway, Australia, the Americas, Europe and ANZ
- **International Network:** Extensive connections with major oil & gas companies and geological surveys globally
- **Technical Excellence:** PhD in Structural Geology from University of Adelaide - strong academic foundation
- **Software Development:** Background in geoscience software development - bridges geology and technology
- **Team Builder:** Proven ability to build and lead technical teams across multiple countries
- **Business Acumen:** Successfully grew Lithodat while maintaining technical excellence - balances business and technical priorities

Dr. Wayne Noble

Chief Information Officer



GDAC-SA Tender Role: Technical Director | 12+ Years Technical Leadership | Scientific Software Developer | Enterprise-Scale Agile Transformation

Technical Director - Agile Leadership & Organizational Change

GDAC-SA Tender Role

Technical Director | 12+ Years Technical Leadership | Scientific Software Developer | Enterprise-Scale Agile Transformation

Current Position

Technical Director

Lithodat Pty Ltd | Melbourne, Australia 2019 - Present (6+ years)

- Technical strategy and platform architecture leadership
- Oversee all technology initiatives and development teams
- Guide technical decision-making and standards
- Lead technical aspects of GDAC-SA tender response
- Mentor and develop technical talent
- Drive innovation in geoscience data management

Core Expertise

- Agile Transformation & Coaching
- Enterprise Software Architecture
- Team Leadership & Development
- Organizational Change Management
- Software Development Lifecycle
- Product Development & Strategy
- DevOps & Continuous Delivery
- Technical Consulting

Key Professional Experience

MYOB - 13 Years 7 Months Total Experience

Agile Coach / Delivery Coach

MYOB | Melbourne, Australia October 2018 - October 2019

- Coached teams and leaders in agile practices and mindset
- Facilitated organizational transformation initiatives
- Developed and delivered training programs
- Helped teams improve velocity and delivery quality
- Supported leadership in adopting agile methodologies

Agile Coach

MYOB | Melbourne, Australia October 2011 - December 2015

- Managed engineering teams delivering enterprise software
- Implemented DevOps practices and continuous delivery
- Drove technical excellence and engineering standards
- Created a model for continuous improvement in teams

Developer / Senior Developer

MYOB | Melbourne, Australia May 2003 - October 2011

- Architected key product features and modules
- Mentored junior and mid-level developers
- Established coding standards and best practices
- Drove adoption of automated testing

Consulting & Advisory Roles

Agile Transformation Consultant

MYER | Melbourne, Australia *November 2019 - April 2020*

- Guided retail technology transformation
- Coached leadership and teams in agile practices
- Facilitated strategic planning sessions
- Helped establish agile centers of excellence

Agile Transformation Consultant

CPA Australia | Melbourne, Australia *February 2018 - October 2018*

- Advised on digital transformation strategy
- Provided guidance on team structure and practices

Agile Transformation Consultant

Australia Post | Melbourne, Australia *October 2017 - February 2018*

- Coached multiple teams across the organization
- Supported leadership in change management

Agile Transformation Consultant

Colonial First State | Melbourne, Australia *March 2017 - July 2017*

- Facilitated quality uplift through adoption of unit testing
- Recommended improvements to delivery practices

Agile Transformation Consultant

Yarra Valley Water | Melbourne, Australia *July 2016 - March 2017*

- Established a tribe cadence of planning
- Coached people new to the role of Product Owners
- Recommended improvements to delivery practices

Agile Transformation Consultant

Hansen Technologies | Melbourne, Australia *January 2016 - July 2016*

- Introduced Agile working
- Trained team members in Scrum practices and rhythms
- Coached the wider business in the changes required to adopt Agile

Education

Graduate Diploma in Software Development (with Distinction)

RMIT University | Melbourne, Australia *2002 - 2006*

- Completed with distinction

PhD - Earth Science

Latrobe University | Melbourne, Australia *1992 - 1997*

- Thermochronology study across eastern Africa
- Highly developed analysis and analytic skills

Bachelor of Science (Honours)

La Trobe University | Melbourne, Australia *1986 - 1990*

- First Class Honours
- Strong foundation in geology and research

Technical Skills

- **Leadership:** Agile coaching, team building, organizational change, strategic planning
- **Software Development:** Full-stack development, architecture design, DevOps practices
- **Methodologies:** Agile, Scrum, Kanban, SAFe, DevOps, Lean
- **Technical:** C#, Java, Python, SQL, cloud platforms (AWS, Azure), CI/CD pipelines
- **Domain Expertise:** Enterprise software, scientific computing, data platforms, Earth Science, thermochronology
- **Tools:** JIRA, Confluence, Git, Jenkins, Docker, Kubernetes

Publications & Speaking

- Multiple conference presentations on agile transformation
- Guest lectures at University of Melbourne on software engineering
- Published articles on technical leadership and team dynamics
- Active contributor to agile and DevOps communities

Relevance to GDAC-SA Tender

- **Technical Leadership:** 12+ years leading technology teams at MYOB - proven ability to deliver complex enterprise platforms
- **Agile Transformation:** Expert in scaling agile practices across organizations - critical for managing distributed teams and stakeholders
- **Enterprise Architecture:** Experience architecting and delivering cloud-based platforms for thousands of users
- **Scientific Software:** Fission Track Studio development demonstrates ability to build specialized scientific data analysis tools for geoscience applications
- **Product Management:** General Manager experience balancing technical excellence with business outcomes and stakeholder needs
- **Team Building:** Grew engineering teams - knows how to scale technical organizations effectively
- **DevOps Excellence:** Implemented modern DevOps practices and continuous delivery - ensures platform reliability and rapid iteration
- **Change Management:** Successfully led organizational transformations at major enterprises (MYOB, Australia Post)
- **Consulting Background:** Broad consulting experience across multiple industries provides flexibility and adaptability
- **Academic Credentials:** PhD in Earth Science demonstrates deep technical knowledge and research capabilities

Keith Dimech

Chief Operating Officer



GDAC-SA Tender Role: Chief Operating Officer | Database Development Expert | 2024 Victorian Infrastructure Award Winner | Industry Leadership (AWA, Beyond Zero)

Chief Operating Officer - Environment and Sustainability Manager

GDAC-SA Tender Role

Chief Operating Officer | Database Development Expert | 2024 Victorian Infrastructure Award Winner | Industry Leadership (AWA, Beyond Zero)

Current Positions

Chief Operating Officer (COO)

Lithodat Pty Ltd | Melbourne, Australia 2019 - Present (6+ years)

- Oversee all operational aspects of the company
- Lead GDAC-SA tender preparation and submission
- Manage client relationships and project delivery
- Database development and data management strategy
- Business development and strategic partnerships
- Financial planning and operational efficiency

Environment and Sustainability Manager

John Holland Water | Melbourne, Australia September 2021 - Present (3+ years)

- Developed and rolled out the Project Pathway to Net Zero Sustainability strategy for John Holland and Melbourne Water, now utilised across multiple projects and contractors
- Environmental and Sustainability Manager for Hobsons Bay Main and Western Treatment Plant RRRC projects, with a combined project value of \$900 million
- Winner of the 2024 Victorian Infrastructure Project Innovation Award - Australian Water Association
- Embedded in the corporate Environment and Sustainability team, undertaking policy writing, tendering, auditing and compliance
- Authorship of the ISO 9001 and 14001 accredited - Environmental, Sustainability and Safety Management Plan / Policy / Procedures
- Member of the HSEQ leadership team, including times as acting Safety Manager
- Ensuring the project achieves and abides by all approvals from various regulators required
- Delivered Australia's First Carbon Neutral Project - led the team to ensure lowest possible carbon emissions onsite through use of new technologies, sustainability through design and detailed materials tracking

Major Awards & Recognition

2024 Victorian Infrastructure Project Innovation Award

Australian Water Association (AWA)

- Recognized for innovative approaches to environmental data management on infrastructure projects
- Implemented novel database systems for tracking environmental compliance

2025 Australian Infrastructure Project Innovation Award

Australian Water Association (Finalist - Announced May 2025)

2025 Built Environment and Infrastructure

Banksia Award (Finalist - Announced April 2025)

Core Expertise

Sustainability Management

- Innovator in the management of digital construction material tracking and carbon accounting, developing a cloud-based SQL database to track material quantities and embodied carbon through project lifecycle
- Proficient in carbon accounting, social expenditure data collection, and regulatory reporting, including SBTi, Building Equality Policy, Indigenous Procurement Policy, and National Greenhouse Energy Reporting Scheme
- Founder of the Beyond Zero – Australian Water Association National Sustainability Specialist Network
- Deep knowledge of the Australian and International Carbon Credit Markets
- Embedding carbon into the cost planning, tendering, and quantity surveying decision-making procedures
- Prior member of the Low Carbon in Concrete Working Group with MTIA; strong understanding of new and emerging low-carbon construction materials, practices, and methods

Environmental Management

- Author of, and ensure compliance with, various environmental management plans, environmental permits, and procedures for cultural heritage, biodiversity, groundwater, contaminated land, noise, air quality and waste
- Extensive experience in approvals management, including EPAV, Council (Native Vegetation, Planning, Heritage Victoria, Traditional Owner Corporations), Parks Vic/DEECA, and Utilities
- Founder of an industry working group, applying for a CRC-P Grant and build a federal relational and geospatial database platform to store, analyse and report on federal environmental contamination and biodiversity data
- Deep knowledge in geotechnical field preparation and execution with a focus on soil and groundwater data collection; expert knowledge in contaminated land field works

Business Corporate Roles

- Corporate strategy, policy and procedure writing. Lead auditor and compliance manager across multiple projects
- Worked on, helped price large Energy, Water, Road and Rail Tenders, including successful bids for Northeast Link, Western Trunk Sewer, Resource Recovery & Reuse Complex, Suburban Rail Loop and Belmont Desalination Plant. Awaiting results on Tarraleah Hydroelectric upgrade, Aurora Treatment Plant and Healesville Treatment Plant
- Management, forecasting and budgeting of multiple million-dollar packages, including environment and sustainability expenses, construction and operation of water treatment plants and spoil cartage and disposal
- Tender, scope, and manage dozens of subcontracts valued between \$10k - \$2m
- 4 Direct reports and in the leadership group of a construction team of 55
- Skilled public speaker and presenter, engaging in industry events, podcasts, community forums, client briefings, stakeholder meetings, toolbox talks, and inductions

Industry Leadership Roles

Committee Member

Australian Water Association (AWA) | Victoria 2024 - 2026

- Victorian Branch Committee Member contributing to industry standards and best practices
- Participate in technical committees on data management
- Networking and knowledge sharing across water industry
- Advocate for innovation in environmental compliance

Founder

Beyond Zero – Australian Water Association National Sustainability Specialist Network 2018 - Present

- Lead sustainability initiatives in construction industry
- Develop frameworks for carbon-neutral infrastructure projects
- Engage with government and industry stakeholders
- Drive adoption of environmental best practices

Lead Applicant

Diagnostic Environmental Reporting Tool – CRC-P Grant 2025

- Applying for federal grant to build relational and geospatial database platform
- Store, analyse and report on federal environmental contamination and biodiversity data
- Establish industry working group for environmental data management

Prior Member

Low Carbon Concrete Working Group – MTIA

- Contributed to development of low-carbon construction materials and practices
- Industry collaboration on sustainability standards

Professional Experience

Senior Environmental Advisor

Clough (Clough Salini Joint Venture) | Snowy Hydro 2.0 Project, NSW *January 2021 - July 2021*

- Senior advisory role with the managing contractor on the \$5.1 billion renewable energy project in NSW being delivered on behalf of the Federal Government and Snowy Hydro Ltd
- Implementation of the Biodiversity Management Plan – balancing construction needs with the conservation of sensitive flora and fauna within the National Park
- Key stakeholder management position – face-to-face coordination with the client (Snowy Hydro), impacted residents and inter-governmental agencies (NPWS, NSW-EPA, DPIE)
- Guide environmental approvals for construction teams to progress works within the Kosciusko National Park (KNP), including road and drainage design, erosion and sediment control planning and approvals, national park clearing permitting and risk mitigation for pollution and sensitive waterways
- Preparing, reviewing, and project managing the implementation of large-scale Erosion and Sediment Control Plans, drainage, and sediment basin construction as per Blue Book specifications
- Managing the ground and surface-water ESDAT chemical database for the Snowy 2.0 Project, required for the upkeep under the Environmental Protection Licence

Environmental Coordinator

John Holland / Bouygues / Lendlease JV | Melbourne Metro Tunnel Project *January 2018 - December 2020*

- Work with key stakeholders including the City of Melbourne, EPA-Vic and Heritage Victoria to ensure planning approval and community consultation prior to staged construction works; ongoing engagement with impacted stakeholders in the CBD including presentations to community reference groups
- Managed environmental monitoring systems, including telemetered noise, air, and dust monitors across CBD sites
- Project managing large groundwater, surface-water and soil contamination monitoring events
- Successful approvals and excavation of twelve Heritage Inventory archaeological digs, the biggest archaeological investigation in Victoria's history

Environment Engineer

WSP | Melbourne, Australia *2016 - 2018*

- Environmental engineering and consulting
- Project management and stakeholder engagement
- Environmental impact assessments and approvals

Environmental Technician

Enqip | Melbourne, Australia *2015 - 2016*

- Environmental monitoring and field work
- Data collection and reporting
- Contaminated land investigations

Geologist

Minerals and Metals Group (MMG) *2011 - 2013*

- Geological field work and exploration
- Data analysis and geological mapping
- Mining industry experience

Education & Professional Development

Executive Master of Business Administration (EMBA)

RMIT University | Melbourne, Australia 2023 - 2025

- Corporate Finance & Corporate Accounting
- Creating Business Strategies & Innovation in a Global Context
- Design Thinking & Marketing
- Ethical Leadership and Business Research Design
- Managing Business Operations & Executive Consulting

Bachelor of Science (Honours) - Geophysics

University of Melbourne | Melbourne, Australia 2014

- Honours degree in Geophysics
- Research and thesis completion

Bachelor of Science - Geology

University of Melbourne | Melbourne, Australia 2009 - 2011

- Major in Geology
- Strong foundation in Earth Sciences

Technical Skills

- **Database Development:** PostgreSQL, SQL, cloud-based database systems, geospatial databases, ESDAT chemical database management
- **Programming:** Python, SQL query optimization, database design, data analysis
- **Environmental Systems:** ISO 9001 and 14001 accredited systems, environmental management plans, sustainability frameworks
- **Data Management:** Construction material tracking, carbon accounting systems, environmental monitoring data
- **Sustainability:** Carbon accounting, embodied carbon tracking, SBTi, National Greenhouse Energy Reporting Scheme
- **GIS & Spatial Analysis:** Geospatial database platforms, spatial data analysis
- **Regulatory Compliance:** EPAV, Heritage Victoria, Traditional Owner approvals, EPA licensing
- **Project Management:** Tender preparation, budget management, subcontract management, team leadership

Certifications & Training

- **ISO 14001 Environmental Management Systems** - Lead Auditor
- **Blue Book** - Erosion and sediment control training
- **Construction White Card**
- **First Aid / Mental Health First Aid Certificate**

Publications & Presentations

- **“Project Pathway to Net Zero”** - Oz Water 2025 Technical Paper
- Multiple industry presentations and conference talks
- Podcasts on sustainability and environmental management
- Community forums and stakeholder briefings

Professional Memberships

- Australian Water Association (AWA) - Victorian Branch Committee Member (2024-2026)
- Environment Institute of Australia & New Zealand (EIANZ)
- Engineers Australia

Referees

James Le - Client Representative +61 429 230 466 Senior Project Manager - Melbourne Water

James Braham - Previous Manager +61 418 915 754 National Environment and Sustainability Manager - Downer

Relevance to GDAC-SA Tender

- **Database Expertise:** Advanced SQL and cloud-based database development skills - directly applicable to building GDAC geological database platform. Proven track record developing innovative environmental data tracking systems
- **Award-Winning Innovation:** 2024 Victorian Infrastructure Innovation Award validates approach to data management and digital systems on large-scale infrastructure projects
- **Full-Stack Experience:** Experience building complete database systems from concept to deployment, including geospatial capabilities
- **Operational Leadership:** COO role at Lithodat demonstrates ability to manage complex operations and deliver on strategic objectives while simultaneously managing major infrastructure projects
- **Dual Role Success:** Successfully managing both Lithodat COO duties and John Holland Environment Manager role shows exceptional time management, organizational skills, and work capacity
- **Major Infrastructure Experience:** Deep understanding of large-scale infrastructure projects (\$900 million combined project value) with government stakeholders and regulatory compliance
- **Tender Experience:** Extensive experience pricing and winning major government tenders (Northeast Link, Western Trunk Sewer, Suburban Rail Loop, etc.) - directly applicable to GDAC-SA tender process
- **Industry Leadership:** AWA Victorian Branch Committee membership, Beyond Zero founder, and CRC-P Grant lead applicant demonstrates thought leadership and industry connections
- **Environmental Data Systems:** Proven expertise in environmental monitoring databases and regulatory reporting systems provides relevant experience for geological survey data management
- **Stakeholder Management:** Exceptional ability to engage with government agencies, regulators, clients (Melbourne Water), and industry stakeholders - critical for GDAC-SA success
- **Executive Education:** Pursuing Executive MBA at RMIT demonstrates commitment to business excellence and strategic thinking
- **Carbon Neutral Delivery:** Led Australia's First Carbon Neutral Project - demonstrates innovation, sustainability leadership, and ability to deliver groundbreaking outcomes
- **Database Innovation:** Developed cloud-based SQL database for construction material tracking and carbon accounting - shows ability to create innovative digital solutions for complex data management challenges

Dr. Mahdi AbuAli, PhD

GDAC Saudi Lead



GDAC-SA Tender Role: Senior Technical Advisor | 35+ Years Saudi Aramco | Fleet Tahreez Managing Director (KSA) | 50+ Prospects Risked | Native Arabic Speaker

Senior Geological Consultant - Saudi Arabia Petroleum Expert

GDAC-SA Tender Role

Senior Technical Advisor | 35+ Years Saudi Aramco | Fleet Tahreez Managing Director (KSA) | 50+ Prospects Risked | Native Arabic Speaker

Current Positions

COO and VP

Raphael Energy Group | Australia (Remote / Australia & KSA) *January 2021 - Present (5 years) | Part-time*

Provide corporate leadership across portfolio strategy, governance cadence, and AOP/budget ownership, working with directors and partners to align technical evaluation with investment decisions. Focused on operating discipline, HSE/ESG, and building digital capability.

Key Achievements: - Established board reporting and rhythm for AOP, performance, and governance - Integrated 3D grav/mag with IHI to prioritize opportunities and move prospects through internal stage-gates - Strengthened stakeholder engagement with directors, investors, and partners

Raphael Energy Group Limited is exploring for oil and gas in the Officer Basin, onshore Western Australia, using an innovative approach integrating 3D gravity/magnetic mapping with microseepage induced indirect hydrocarbon indications (IHIs) such as radiometric gamma anomalies, to provide 3D GravMag IHI leads.

Managing Director

Fleet Tahreez Company Limited | Riyadh, Saudi Arabia *July 2025 - November 2025 (5 months) | Full-time*

Leading mineral exploration initiatives that apply satellite- and AI-enabled geoscience to generate, rank, and de-risk targets for tier-one discoveries. Bridge operator needs with advanced sensing, turning technical insight into decision inputs for work-program planning and stakeholder reviews across KSA.

Key Achievements: - Introduced satellite/AI workflows that improved speed of exploration, target generation and prioritization - Delivered decision inputs adopted in planning reviews across KSA

Senior Advisor

Axiom Exploration Group | Saskatchewan, Canada (Hybrid) *April 2021 - June 2025 (4 years 3 months) | Part-time*

Expanded remit from hydrocarbons to minerals, reflecting demand and the energy transition.

Key Achievements: - Supported opportunity screening and BD engagements for regional mandates - Broadened scope to minerals exploration aligned with market priorities

Recent Consulting Experience

Consultant & Project Manager

Halliburton Kuwait | Kuwait *February 2023 - October 2023 (9 months) | Full-time*

Led a multidisciplinary team on Paleozoic/Triassic gas exploration, delivering a ranked prospect inventory with full risking and volumetrics to inform client decision-making and program planning.

Key Achievements: - Matured, ranked, and risked 10 prospects across ~3,000 km² with supporting volumetrics - Inventory adopted for the next stage of planning by KOC stakeholders

Saudi Aramco - 35 Years 9 Months of Distinguished Service

April 1985 - December 2020

Senior Geological Consultant

Saudi Aramco - Upstream Advanced Research Center | Dhahran, Eastern, Saudi Arabia *June 2014 - December 2020 (6 years 7 months) | Full-time*

Drove deployment of exploration technologies to lower risk, mentored junior geoscientists, and embedded new methods into daily workflows—strengthening technical quality and decision confidence. Published papers and authored patents.

Key Achievements: - Embedded risk-reducing technologies and practices across teams - Delivered internal courses and editorial leadership to raise technical standards

Senior Geological Consultant

Saudi Aramco - Strategic Planning & Optimization (Exploration) | Dhahran, Saudi Arabia *February 2011 - June 2014 (3 years 5 months) | Full-time*

Member of a senior team standardizing portfolio risking and shaping the exploration drilling schedule. Partnered with leadership on investment cases, approvals, and AOP delivery.

Key Achievements: - Standardized prospect risking across the national portfolio; matured and risked 50+ prospects, several drilled and resulting in oil & gas discoveries - Prepared decision papers and presented to senior leadership to inform approvals and execution

Senior Geological Consultant

Saudi Aramco - Upstream Ventures (JV Programme Manager) | Dhahran, Saudi Arabia *September 2006 - February 2011 (4 years 6 months) | Full-time*

Managed JV exploration programs (SRAK, EniRepSa, SSG, LukSar), representing Aramco in governance forums and partner committees. Oversight spanned large onshore areas with planned wells and development concepts progressed with JV partners.

Key Achievements: - Oversaw >200,000 km² of JV acreage; planned several exploration wells and advanced the Kidan Arab sour-gas development plan - Prepared decision papers and secured partner approvals for work programs and budgets

Senior Geologist - Earlier Roles

Saudi Aramco | Dhahran, Saudi Arabia *June 1989 - September 2006 (17 years 4 months) | Full-time*

Progression: Geochemist → Reservoir/Exploration Geologist → Geological Specialist/Team Lead

Progressed from hands-on geochemistry into exploration and specialist leadership, supporting prospect generation through petroleum systems modeling, source–oil correlations, reservoir characterization (clastic & carbonate), and 2D/3D seismic interpretation. Partnered with multi-disciplinary teams to translate subsurface insight into drillable opportunities and helped establish repeatable workflows adopted across exploration.

Key Achievements: - Built regional play-fairway and charge/migration frameworks used in exploration interpretation - Delivered geochemical correlations and subsurface interpretations that underpinned prospect maturation and investment decisions

Geologist

Saudi Aramco | Dhahran, Saudi Arabia *April 1985 - September 1996 (11 years 6 months) | Full-time*

Geologist and Geochemist, worked on Exploration of Central Arabia, Northwest and Red Sea areas. Conducted R&D studies on the source rock and migration of hydrocarbons, established a new source rock for the Paleozoic of Saudi Arabia. Published and presented many articles on the origin of hydrocarbons in the Paleozoic petroleum systems.

Core Expertise

- **Regional Geology:** Deep expertise in Arabian Peninsula, Middle East petroleum systems
- **Exploration Strategy:** Portfolio management, prospect risking, resource assessment
- **Basin Analysis:** Petroleum systems modeling, play concept development
- **Leadership:** Team building, mentoring, stakeholder management, multicultural team leadership
- **Saudi Operations:** 35+ years working in Saudi Arabia with comprehensive understanding of local context
- **Technical Integration:** Multidisciplinary data integration and interpretation
- **Business Development:** Partnership negotiations, JV management, joint ventures
- **Emerging Technologies:** Satellite/AI-enabled exploration, 3D GravMag IHI methods
- **Minerals Exploration:** Recent experience expanding from hydrocarbons to critical minerals

Technical Skills

- **Exploration Management:** Project management, exploration portfolio analysis, budgeting, strategic planning
- **Geological Analysis:** Basin analysis, petroleum systems analysis, reservoir characterization, field work, sampling
- **Data & Analytics:** Data analysis, quantitative research, qualitative research, analytical skills
- **Technical Disciplines:** Seismic interpretation, log interpretation, geochemistry, source rock analysis
- **Leadership & Management:** Team management, coaching, multicultural team leadership, contractual obligations
- **Specialized:** AI exploration, satellite-enabled exploration, mineral exploration, HSE/ESG

Education

PhD - Geosciences

RWTH Aachen University | Germany *2001 - 2005*

- Advanced research in petroleum systems and basin analysis
- Published research in peer-reviewed journals
- Activities and societies: AAPG, SPE, EAGE, IMOG, ALAGO, AAPG

Master of Science (MS) - Geochemistry

Colorado School of Mines | Colorado, USA *January 1986 - May 1989*

- **Grade: 3.2/4.0**
- Specialization in petroleum geochemistry

Bachelor of Applied Science (BASc) - Industrial Chemistry

King Fahd University of Petroleum & Minerals (KFUPM) | Dhahran, Saudi Arabia *September 1980 - January 1985*

- **Grade: 3.0/4.0**
- Strong foundation in chemistry and petroleum sciences

Languages

- **Arabic:** Native speaker - critical for Saudi stakeholder engagement
- **English:** Fluent - international business and technical communication

Professional Memberships

- American Association of Petroleum Geologists (AAPG)
- Society of Petroleum Engineers (SPE)
- European Association of Geoscientists and Engineers (EAGE)

Relevance to GDAC-SA Tender

- **Saudi Arabia Expertise:** 35+ years 9 months working in Saudi Arabia at Saudi Aramco (1985-2020) - unparalleled understanding of local geology, culture, and stakeholder landscape
- **Recent Saudi Experience:** Managing Director at Fleet Tahreez (Jul-Nov 2025) in Riyadh - current, active relationships with Saudi geological authorities and stakeholders
- **Arabic Language:** Native speaker enables direct, nuanced communication with Saudi Geological Survey and government stakeholders
- **Senior Advisory Capability:** Current COO/VP at Raphael Energy and recent Senior Advisor roles demonstrate continued relevance and thought leadership
- **Proven Leadership:** Led major strategic initiatives at Saudi Aramco managing >200,000 km² and risking 50+ prospects with multiple discoveries
- **Resource Assessment:** Deep expertise in probabilistic risking and resource estimation - directly applicable to GDAC database requirements
- **International Collaboration:** Extensive JV experience with major oil companies (ENI, Lukoil, Shell) - knows how to work with diverse stakeholders and partners
- **Data Integration:** Career focused on integrating multidisciplinary geoscience data - core requirement for GDAC platform development
- **Mentoring:** Decades of experience training and developing geoscientists - can support knowledge transfer to Saudi teams
- **Gulf Region Network:** Extensive professional network across Saudi Arabia and broader Gulf region from 35+ years of service
- **Innovation Focus:** Recent experience with satellite/AI-enabled exploration and 3D GravMag IHI methods demonstrates comfort with emerging technologies
- **Minerals Expertise:** Expanded scope to minerals exploration aligns with Saudi Vision 2030 diversification beyond hydrocarbons
- **Current Saudi Presence:** Recent 5-month assignment in Riyadh (2025) demonstrates ability to work on-ground in KSA

Dr. Qusay Abeed, PhD

Geological Director



GDAC-SA Tender Role: GDAC Technical Director | Native Arabic Speaker | Halliburton Petroleum Systems Lead | CCUS Expertise (Vision 2030 Alignment)

GDAC Technical Director - Petroleum Geochemist

GDAC-SA Tender Role

GDAC Technical Director | Native Arabic Speaker | Halliburton Petroleum Systems Lead | CCUS Expertise (Vision 2030 Alignment)

Current Positions

Petroleum Geochemist Senior / Permedia and DSG Petroleum Systems Lead QA

Halliburton | United Kingdom *March 2020 - Present (5 years)*

- Senior role overseeing petroleum systems quality assurance for Permedia and DSG products
- Expertise in C++ programming and Carbon Capture, Utilisation and Storage (CCUS)
- Technical leadership for petroleum systems modeling and interpretation

Permedia CO2 Team Agile Scrum Master

Landmark Software (Halliburton) | Oxford, England, United Kingdom (Hybrid) *March 2017 - Present (8 years)*

- Dealing with & developing technologies for basin modelling and CO2 storage
- Leading Agile/Scrum practices for CO2 technology development team
- Technical expertise in C++ and Carbon Capture, Utilisation and Storage (CCUS)

Core Expertise

- Carbon Capture & Storage (CCUS)
- Petroleum Geochemistry
- Basin Modelling
- Petroleum Systems Analysis
- Source Rock Characterization
- Oil-Source Rock Correlation
- C++ Programming
- Agile/Scrum Project Management

Languages

- **Arabic:** Native speaker - Direct communication with Saudi Geological Survey stakeholders
- **English:** Fluent - International professional communication

Key Professional Experience

Petroleum Geochemist / Petroleum Systems Lead

Exploration Insights, Landmark, Halliburton | Abingdon, UK *October 2013 - March 2020 (6 yrs 6 mos)*

- Working as Petroleum System Lead, dealing with search, gathering, and processing petroleum systems data
- Global coverage: South America, Central America, North Africa, Lower 48 (USA) and Middle East
- **Created Global Petroleum Systems Interpretations Maps** for diverse regions
- Deep understanding of Middle East petroleum systems including Mesopotamian Basin
- Skills: Log Interpretation, Research and Development (R&D)

Petroleum Geologist / Post-Doctoral Researcher

RWTH Aachen University | Aachen, Germany *October 2012 - October 2013 (1 yr 1 mo)*

- Post-doctoral research on source rock/oil characterization, oil-source rock correlation, and oil-oil correlation
- Focus on bulk properties, biomarkers and molecular geochemistry of oil
- **Basin modelling for the Mesopotamian Basin of Iraq** - one of the largest petroleum systems in the world
- Gained experience in managing petroleum system modelling and seismic interpretation

Research Associate

RWTH Aachen University | Aachen, Germany *October 2008 - October 2012 (4 yrs 1 mo)*

- Petroleum systems research at Institute of Geology and Geochemistry of Petroleum and Coal
- Advanced research in organic geochemistry and thermal/burial history
- Skills: Research and Development (R&D), Presentations

Researcher and Exploration Instructor

Tikrit University | Tikrit, Iraq *November 2006 - May 2008 (1 yr 7 mos)*

- Researcher at the Institute of Geology and Geochemistry of Petroleum and Coal at RWTH Aachen University
- Main field of interest: Organic geochemistry of petroleum and source rocks
- Thermal/burial history and basin modelling research
- Skills: Research and Development (R&D), Presentations

Education

PhD - Petroleum Geochemistry & Basin Modelling

RWTH Aachen University | Aachen, Germany *2008 - 2012*

- **Grade: Magna Cum Laude**
- Thesis focused on Mesopotamian Basin petroleum systems

MSc - Petroleum Geology

University of Baghdad | Baghdad, Iraq *2002 - 2006*

- **Grade: 81.91%**

BSc - Geology/Earth Science

University of Baghdad | Baghdad, Iraq *1998 - 2002*

- **Grade: 71.7%**

Technical Skills

- **CCUS Technologies:** Carbon capture, utilisation and storage systems - aligns with Saudi Vision 2030 sustainability initiatives
- **Programming:** C++ for petroleum systems modeling and basin simulation
- **Geochemical Analysis:** Biomarkers, molecular geochemistry, source rock characterization
- **Basin Modelling:** Thermal history, burial history, petroleum systems modeling
- **Seismic Interpretation:** 2D/3D seismic data interpretation
- **Log Interpretation:** Well log analysis and correlation
- **Project Management:** Agile/Scrum methodologies

Geographic Expertise

- **Middle East:** Mesopotamian Basin (Iraq), North Africa - deep regional knowledge
- **Americas:** South America, Central America, Lower 48 (USA)
- **Europe:** UK, Germany - extensive work experience

Relevance to GDAC-SA Tender

- **Native Arabic Speaker:** Direct, fluent communication with Saudi Geological Survey stakeholders and government officials
- **Middle East Geology Expertise:** Deep understanding of regional petroleum systems including Mesopotamian Basin research
- **Halliburton Connection:** Former colleague of Dr. Fabian Kohlmann with shared company culture and work methodology
- **CCUS Knowledge:** Expertise in carbon capture and storage aligns perfectly with Saudi Arabia’s sustainability goals and Vision 2030 initiatives
- **Global Petroleum Systems Experience:** Created interpretations for diverse regions including Middle East, North Africa, and Americas
- **Basin Modelling Expertise:** Essential technical capability for geological data platform development and analysis
- **Programming Skills:** C++ expertise valuable for database development and system integration
- **Academic Excellence:** PhD Magna Cum Laude from prestigious RWTH Aachen University
- **Project Management:** Agile Scrum Master experience for effective team coordination
- **Research Background:** Strong R&D capabilities from academic and industry research roles

Moritz Theile

Chief Technology Officer



GDAC-SA Tender Role: Chief Technology Officer | 6+ Years Lithodat CTO | Full-Stack Architect

Chief Technology Officer

GDAC-SA Tender Role

Chief Technology Officer | 6+ Years Lithodat CTO | Full-Stack Architect

Current Positions

Chief Technology Officer (CTO)

Lithodat Pty Ltd | Melbourne, Australia 2019 - Present (6+ years)

- Architect and Lead Developer for all Lithodat technical infrastructure
- Building IT Infrastructure from ground up
- Developer Onboarding and Team Building
- Responsible for all technical architecture and system design decisions

Self-Employed Software Architect

MTheile.com | Greater Munich Metropolitan Area, Germany 2011 - Present (14 years)

- Delivering IT solutions that actually work
- Starting with analysis, ending with thought-through systems
- Multiple successful project deliveries across diverse industries

Core Expertise

- Full-Stack Development
- Cloud Architecture (AWS)
- Java / Spring Boot
- JavaScript / TypeScript / React
- SQL / Database Design
- Data Analytics (Tableau, ELK)
- Agile Methodologies
- System Architecture

Key Projects at MTheile.com

HotelOrga

Hotel Management Platform

- Comprehensive platform for hotel operations management
- Full-stack development with modern architecture

TeamTable

Generic Database System

- Flexible database system for team collaboration
- Custom data modeling and analysis capabilities

MonaStack

Technology Stack Development

- Modern technology stack for rapid application development
- Reusable components and architecture patterns

Cutworks

Laser-Cut Metal Parts Ordering System

- E-commerce platform for manufacturing industry
- Integration with manufacturing systems and logistics

Docsafe

Consulting and Development for Swisscom

- Enterprise-level document management solution
- Security and compliance requirements

Professional Experience

Project Manager

4Soft | Germany 2007 - 2011 (4 years)

Led multiple enterprise software projects for financial services industry:

DynSBatch

Bulk Data Processing Software

- High-performance batch processing system
- Enterprise-scale data transformation and validation

BeteiligungsIntegrator

Interface for Investment Funds

- Integration platform for financial institutions
- Complex data synchronization and validation

DWH4OSPlus

Framework for Data Analysis

- Data warehouse framework for business intelligence
- Analytics and reporting capabilities

WAC Manager

Portfolio Management Software

- Investment portfolio tracking and analysis
- Real-time market data integration

Java Developer

Macquarie University | Sydney, Australia 2005 - 2006 (1 year)

Developer for Australian Higher Education systems:

MAMS (Meta Access Management System)

Australian Higher Education Platform

- Identity and access management for university systems
- Integration across multiple institutions

Autograph

Shibboleth Extension

- Enhanced authentication capabilities
- Single sign-on infrastructure

Education

Computational Science

Technical University of Munich (TUM) | Munich, Germany 1998 - 2004

- Focus on computational methods and software engineering
- Strong foundation in mathematics and computer science

Exchange Student

Hoover Highschool | Fresno, California, USA 1994 - 1995

- International education experience

Highschool Graduation (Abitur)

Adalbert-Stifter Gymnasium | Germany 1987 - 1997

Technical Skills

- **Backend Development:** Java, Spring Boot, SQL, Maven, REST APIs
- **Frontend Development:** JavaScript, TypeScript, React, HTML5, CSS3
- **Cloud & Infrastructure:** AWS (EC2, S3, RDS, Lambda), Docker, CI/CD
- **Database:** SQL (PostgreSQL, MySQL), NoSQL, Data Modeling
- **Data Analytics:** Tableau, ELK Stack, UML, Data Visualization
- **Methodologies:** Agile, Scrum, Kanban, DevOps practices
- **Architecture:** Microservices, Distributed Systems, System Design

Professional Skills

- **Leadership:** Team building, mentoring, technical direction
- **Problem Solving:** Complex system debugging and optimization
- **Communication:** Technical documentation, stakeholder management
- **Creativity:** Innovative solutions to technical challenges

Relevance to GDAC-SA Tender

- **CTO & Lead Architect:** 6+ years as Chief Technology Officer at Lithodat - responsible for all technical infrastructure
- **Full-Stack Expertise:** Comprehensive skills across backend (Java, Spring Boot, SQL) and frontend (React, TypeScript) development
- **Cloud & Infrastructure:** AWS experience essential for scalable geoscience data platform deployment
- **German Engineering Excellence:** Technical University of Munich (TUM) Computational Science education
- **International Experience:** Work experience in Germany, Australia, and Switzerland
- **Project Management:** 4 years leading complex software projects at 4Soft including data processing and analytics
- **Entrepreneurial:** Successfully running independent software architecture consultancy since 2011
- **Data Analytics:** Experience with data modeling, Tableau, ELK Stack - crucial for geoscience data platform
- **Higher Education Systems:** Experience building systems for Macquarie University (MAMS, Autograph)

Dr. Behnam Sadeghi

ML Technical Advisor



GDAC-SA Tender Role: Technical Advisor | First Australian IAMG Vistelius Award | Fulbright Scholar | 58+ Publications | 3 Python Packages | JORC Certified

Principal Consultant & Research Fellow PhD, FAIG, FAusIMM, FAAG, RPGeo, Fulbrighter

GDAC-SA Tender Role

Technical Advisor | First Australian IAMG Vistelius Award | Fulbright Scholar | 58+ Publications | 3 Python Packages | JORC Certified

Major Awards & Recognition

- **2023 IAMG Vistelius Award** - First Australian recipient
- **2022-2024 Fulbright Scholar** (USD 29,325)
- **2020 IAMG Founders Award** - First Ever
- **FAusIMM** (Fellow of Australasian Institute of Mining and Metallurgy)
- **FAAG** (Fellow of Association of Applied Geochemists)
- **FAIG** (Fellow of Australian Institute of Geoscientists)
- **RPGeo 10295** (Registered Professional Geoscientist)

Current Positions

Research Fellow - Critical Minerals Exploration

CSIRO (Australian Resources Research Centre) | Australia 2023 - Present

- Leading research on critical minerals exploration methodologies
- Focus on rubidium, lithium, and rare earth elements

Visiting Research Fellow

UNSW - Dept. of Biological, Earth and Environmental Sciences 2022 - Present

Principal Consultant

Independent Practice 2022 - Present

Research Fellowships

- **Fulbright Research Scholar** - Carnegie Institution for Science, Earth and Planets Laboratory + Stanford University, USA (2023-2024)
- **Honorary Research Fellow** - EarthByte Group, University of Sydney (2022-2023)

Core Expertise

- Fractals & Multifractals
- Exploration Geochemistry
- Uncertainty Quantification
- Machine Learning in Geosciences
- GIS & Data Analytics
- Critical Minerals
- Python Development
- JORC Code Reporting

Publications Summary

- **58+ peer-reviewed journal articles** in leading geoscience journals
- **34+ conference presentations, invited lectures and keynote speeches** (oral and poster)
- **2 books** including Elsevier reference book on Fractals and Multifractals
- **8 book chapters** in Encyclopedia of Mathematical Geosciences
- **3 Python packages** developed for geoscience applications

Key Publications

“Fractals and Multifractals in the Geosciences”

Elsevier Reference Book, 2024

- Comprehensive reference work on fractal applications in geosciences
- ISBN: 978-0-323-90897-9

Selected Journal Articles

- Sadeghi, B., Cohen, D.R. (2023) “Decision-making within geochemical exploration data based on spatial uncertainty” - Ore Geology Reviews
- Sadeghi, B. (2022) “Chatterjee Correlation Coefficient: A robust alternative for classic correlation methods” - Ore Geology Reviews
- Sadeghi, B. (2021) “Simulated-multifractal models: a futuristic review” - Ore Geology Reviews

Python Packages Developed

- **pyClusterWise** - Clustering method selection with uncertainty analysis
- **IRWPy** - Inverse Radius Weighting interpolation for spatial data
- **TripleCpy** - Chatterjee Correlation Coefficient implementation

Editorial & Leadership Roles

Associate Editor (3 journals)

- Ore Geology Reviews
- Geochemistry/Chemie der Erde
- Ore and Energy Resource Geology

Editorial Board Member (5 journals)

- Applied Computing and Geosciences
- Earth Science Informatics
- Minerals
- Mining
- Discover Minerals

Professional Society Leadership

- AAG Councillor (2024-2025)
- AAG Strategic Planning Team (2017-2027)
- IAMG Outreach Committee Member (2023-Present)
- IAMG Liaison Committee Member (2023-Present)
- AIG Official Assessor for RPGeo applications

Key Professional Experience (2007-Present)

Senior Geologist & Consultant

Perpetual Resources Ltd (ASX: PEC) | Australia 2022

Senior Geologist & Data Analyst

Impact Minerals Ltd (ASX: IPT) | Australia 2021 - 2022

Researcher / Technical Software Supporter

EarthByte Group, University of Sydney | Australia 2020 - 2021

- GPlates software tutorials and technical support

Senior Geologist & Data Analyst

Satrap Resources (WA) | Western Australia 2020 - 2021

Senior Geologist

Hampa Behineh Consultant | Iran 2014 - 2020 (6 years)

Researcher

James Cook University - Economic Geology Research Unit (EGRU) | Australia 2015 - 2017

Education

Ph.D. of Geosciences

UNSW, Australia 2018 - 2020

- Thesis: “Quantification of Uncertainty in Geochemical Anomalies in Mineral Exploration”

Master of Mining Engineering - Exploration

Islamic Azad University (IAU), Iran 2009 - 2012

Bachelor of Mining Engineering - Exploration

University of Kurdistan (UoK), Iran 2004 - 2009

Recent Certifications (2024-2025)

- Learn to Lead with UNSW 2025: Strategic Thinking for Impactful Leadership
- Standard Mental Health First Aid - Australian Red Cross
- **Professional Certificate in JORC Code Reporting - AusIMM**
- Using GenAI Effectively at Work - CSIRO
- Python for Geoscientists - Curtin University
- Regolith Geology and Geochemistry - CSIRO
- PMP Certification Exam Prep Course - Udemy

Teaching & Mentoring

- IAMG Short Course “Applied Geochemistry and Analytics” (August 2023)
- GEOL 5806 Graduate Course in Exploration Geochemistry - Laurentian University (December 2022)
- Co-advisor on Geology-Machine Learning projects - UNSW (2021-Present)
- GPlates software tutorials provider - EarthByte Group (2020-2021)

Professional Society Memberships

- Australian Fulbright Alumni Association (AFAA)
- The American Geophysical Union (AGU)
- The Geological Society of America (GSA)
- AusIMM - Fellow (FAusIMM)
- AIG - Fellow and Registered Professional Geoscientist (FAIG, RPGeo)
- European Geosciences Union (EGU)
- Society for Geology Applied to Mineral Deposits (SGA)
- IAMG - Lifetime Member
- AAG - Fellow (FAAG)

Relevance to GDAC-SA Tender

- **Data Science Expertise:** Extensive experience in computational geochemistry and advanced data analytics
- **Critical Minerals Focus:** Current CSIRO research on critical minerals exploration directly aligns with resource assessment needs
- **International Collaboration:** Widespread network with geological surveys (Australia, Sweden, Cyprus, Iran, North Ireland, India, Canada, Norway) and research/industry institutes globally, e.g., Australia, Kazakhstan, Turkey, the US, Canada, Spain, China, Germany, Italy, Nigeria, Morocco, South Africa, the UK, Czech Republic, etc.
- **Software Development:** Proven ability to develop practical tools (3 Python packages) for geoscientific applications
- **Industry + Academia Bridge:** Unique combination of senior industry experience and academic research credentials
- **JORC Competence:** Professional Certificate in JORC Code Reporting (2025) - essential for resource estimation
- **Award Recognition:** First Australian recipient of IAMG Vistelius Award demonstrates international standing
- **Publications:** 58+ peer-reviewed articles and Elsevier reference book establish thought leadership
- **Editorial Experience:** Associate Editor for 3 major journals provides quality assurance expertise
- **Teaching Capability:** Graduate-level course instruction and mentoring experience
- **Fulbright Scholar:** Prestigious international research fellowship at Carnegie Institution and Stanford University
- **Machine Learning:** Co-advisor on geology-ML projects brings AI/ML capabilities to geoscience data platforms

Vinko Novak

Head of Data Security

Digital Transformation | Domain-Driven Design | Enterprise Architecture
+49 151 11 44 68 77 | vinko.novak@scenaryo.de



GDAC-SA Tender Role: Strategic Advisor | 18+ Years Digital Transformation | Deutsche Bahn/BMW/GEMA Architecture Reviews | DDD Expert

Professional Summary

Experienced management consultant specializing in the development of digital solutions. Expertise ranges from innovative methods like Domain-Driven Design and Storming workshops to the practical implementation of digital platforms. Throughout career, conducted important architecture assessments for well-known companies such as Deutsche Bahn, BMW, and GEMA to critically evaluate systems and enable future-oriented strategic planning.

Current Position

Management Consultant, Co-Founder & CEO

Scenaryo GmbH | Munich, Germany

April 2017 - Present (8+ years)

- Actively seek innovative projects and work successfully as management consultant to drive innovation in digital landscape
- Lead strategic consulting engagements for major German enterprises
- Conduct architecture reviews and assessments for critical systems
- Facilitate Domain-Driven Design and Event Storming workshops
- Guide C-level executives on digital transformation strategies

Core Expertise

- | | |
|--------------------------------------|-----------------------------------|
| • Domain-Driven Design (DDD) | • Event Storming & Game Storming |
| • Enterprise Architecture Reviews | • Digital Transformation Strategy |
| • Agile Leadership (Scrum/Kanban) | • Tech Lead & Software Architect |
| • Cloud & Microservices (K8s/Docker) | • Project Turnaround & Recovery |

Key Experience Metrics

- **Tech Lead:** >10 years leading development teams and technical initiatives
- **Software Architect:** >10 years designing enterprise systems and platforms
- **Consultant:** >10 years advising organizations on digital transformation
- **Agile Roles:** >5 years as Scrum Master and Product Owner

Selected Major Projects

CDW, Paderborn (Transportation) - Startup to Funding Round

June 2023 - March 2024

- Consolidated startup strategy within CDW using Game Storming and Event Storming (3 days)
- Defined software architecture and completed business model based on new strategy
- Convinced investors in first funding round with architecture and business model

NDA Client, Munich (Media) - GET On System Integration

January 2024 - Present

- Project manager and technical consultant at program level for core system integration
- Responsible for technical integration and migration to service provider
- Visualize complex business and technical relationships in large workshops (>30 participants)
- Developed fast and efficient proof-of-concept prototypes for quality assessment
- Creating integration concept with enterprise and software architects from all participating companies

GEMA / IT4IPM, Munich (Media) - SOLIS Licensing

January 2022 - April 2023

- Software Architect leading team of 10 developers in SOLIS project
- Made and enforced architectural decisions for licensing platform
- Applied creative methods to visualize architecture for C-level management
- Translated complex technical concepts into understandable visualizations
- Enabled effective strategy that contributed to project success

ACTINEO, Cologne (Insurance) - DDD Strategy

January 2021 - July 2022

- Management consultant for company reorganization aligned with future architecture
- Led interactive Domain-Driven Design workshops (online delivery)
- Significant progress visualizing domains and company's new strategic direction
- Advised DevOps lead and software architects on strategy implementation

Wertkontor Wolf KG, Heidelberg (Insurance) - HR-Benefit Pension Calculator

September 2022 - Present

- Took over and realigned definition and development of Rentenrechner digital platform
- Platform designed for marketing employee benefits and insurance
- Integrated various insurance APIs into common application
- Developed easy-to-use user interface simplifying sales process for distribution partners

NDA Client, Munich - GDPR Deletion Concept

January 2020 - December 2020

- Worked with legal department to develop data protection and deletion concept
- Introduced concept across all directorates through collaboration
- Created effective and legally valid approach to implementing GDPR provisions
- Serves as reference work for developing GDPR-compliant systems in internal landscape

GEMA / IT4IPM, Munich (Media) - Systems Integration Program

June 2015 - December 2019 (4.5 years)

- Tech Lead, Business Analyst, Software Architect, Product Owner, and Tester
- Successful integration of master data in microservice landscape
- Cooperative project of three leading European collecting societies
- Analyzed requirements, defined technical integration architecture
- Created interfaces between societies and conducted comprehensive automated tests
- Advocated for customer interests in joint venture

Deutsche Bahn, Frankfurt (Transportation) - System Audit

April 2015 - June 2016

- System audit to improve maintainability and extensibility of quality characteristics and delays system
- Auditor, software architect, and agile trainer roles
- Interviewed teams, performed static code analyses and code reviews
- Implemented Kanban and adapted development processes
- Application still in productive use beyond planned shutdown date due to action catalog

Deutsche Bahn, Frankfurt (Transportation) - PIA Architecture Review

August 2019 - September 2019

- Architecture review of employee application PIA for Deutsche Bahn
- Graphical representation decisive in evaluating system and realigning project
- PIA successfully developed further based on understanding created
- Application planned for deployment at all German train stations

Previous Professional Experience

Senior Consultant

Cassini AG | Germany

July 2013 - March 2017 (3 yrs 9 mos)

- Led projects and coordinated teams to develop and deliver innovative software systems
- Applied consulting skills to guide clients through digital transformation

Software Architect

4Soft GmbH | Germany

March 2006 - July 2013 (7 yrs 5 mos)

- Experienced software developer with strong technical skills
- Successfully applied expertise in development of custom software

Skills & Methodologies

Methodologies & Practices

- Strategic and tactical Domain-Driven Design (DDD)
- Event Storming, Game Storming workshops
- Design Thinking, Design Sprint facilitation
- Object-Oriented Analysis, Design and Programming (OOA, OOD, OOP)
- Test Driven Development (TDD)
- Clean Code, Design Patterns
- Coaching, Architecture Reviews
- Project Turnaround expertise
- UX Design
- Lean/Kanban, SCRUM

Technology Stack

- **Cloud & Infrastructure:** Kubernetes, Docker, Terraform
- **Programming:** Java, Python, SQL
- **Workflow/BPM:** BPMN (Camunda)
- **Databases:** CouchDB, relational databases
- **Other:** Data Pipelines, Automation, Microservices architecture

Languages & International Experience

- **German:** Native speaker
- **English:** Fluent - International business communication
- **Croatian:** Fluent

Experience working with international teams from Germany, England, America, Tunisia, Hungary, and Romania.

Education

Diplom Informatiker (Master's equivalent in Computer Science)

Technical University of Munich (TUM) | Munich, Germany

2000 - 2005

- Comprehensive computer science education from prestigious German technical university
- Focus on software engineering, distributed systems, and architecture

Relevance to GDAC-SA Tender

- **Digital Transformation Expertise:** 18+ years leading digital platform development and modernization initiatives
- **Strategic Consulting:** Proven ability to align technology with business strategy at C-level (GEMA, Deutsche Bahn, BMW)
- **Domain-Driven Design:** Expert in DDD, Event Storming, and strategic architecture - essential for complex geological data platform design
- **Enterprise Architecture:** Architecture reviews for major organizations (Deutsche Bahn, BMW, GEMA) - brings enterprise-scale expertise
- **Agile Leadership:** 10+ years as Tech Lead with Scrum Master and Product Owner experience - effective team coordination
- **Complex Integrations:** Successfully integrated systems across multiple organizations in joint ventures - critical for GDAC data platform
- **Visualization Skills:** Ability to translate complex technical concepts for executive decision-making - bridges technical and business stakeholders
- **International Experience:** Collaboration with teams across Germany, UK, USA, Tunisia, Hungary, Romania - global perspective
- **Startup & Innovation:** Experience taking startups from concept to successful funding rounds - entrepreneurial mindset
- **Cloud & Modern Tech:** Expertise in Kubernetes, Docker, Terraform, microservices - modern infrastructure capabilities for scalable platform

Annemarie Grass

Head of Cyber Defence

Email: annemariegrass123@outlook.com

An experienced leader with over 10 years leading both operational and technical teams across cyber domains. Specialist in cyber incident response, threat hunting, and implementing detection and response capabilities. I am passionate about building, operating, and enhancing organisations defensive security, responding to security incidents, and uplifting overall security maturity.

Professional Experience

- Head of Cyber Defence | IDP Education Pty Ltd | June 2023
- Senior Manager, Defensive Security | PricewaterhouseCoopers (PwC) | July 2022
- Manager, Defensive Security | PwC | June 2021
- Cybersecurity Lead Forensics & Information Intelligence | AGL | January 2020
- Cybersecurity Lead 3rd Party Security Management | AGL | October 2018
- Lead Financial Crime AML/Sanctions | NAB | May 2018
- Senior Business Analyst | NAB | March 2016
- Senior Change Analyst September | NAB | September 2015

Training and Education

- Foundations of Directorship | Australian Institute of Company Directors | 2024
- Charles Sturt University | Master of Information Technology; Cybersecurity | 2020
- Nuix | Workstation Data Discovery Specialist | 2019
- Adaptive Frameworks | PRINCE 2 – Foundation and Practitioner | 2014
- Griffith University | Bachelor of Business; Mgr. Management and Human Resources | 2013

Demonstrated Skills and Strengths

- Ability to effectively bridge the gap between heavily technical environment and business strategy requirements to deliver outcomes to an organisation, achieved through balancing communication styles across stakeholders at all levels.
- Collaborative approach to problem solving, with ability to engage teams and bring them on the journey to achieve outcomes for organisations.
- Experienced Incident Responder coordinating activities for major multi-national organisations and global operations. Routinely manages multiple teams of investigators spanning across technology, internal audit and business assurance functions, profiles threat actors, works closely with senior executives on sensitive investigations and leading meaningful Post Incident Reviews.

Employment History

IDP Education Ltd
Head of Cyber Defense
Dimensions and Key Achievements

- Responsible for the establishment of the Security Operations Centre, including development of technical incident response plans, playbooks and training and onboarding of new security operations analyst.
- Responsible for leading the establishment of a Cyber Deliver Centre in Manila from initial market scan, vendor partnering and onboarding of 18 experienced cyber security practitioners to support IDP in delivery of it's Cyber Uplift Strategy.
- Responsible for the establishment of a Global Crisis Management Framework for IDP Education, working and engaging with senior executives to develop and refine the framework.
- Accountable for the delivery of several global incident response activities, working with regional technology teams to improve detection, response and recovery capabilities for regional technology teams responsible for local technology infrastructure.
- Led the global rollout of new Endpoint Detection and Response toolset for Servers and Workstations.
- Development of detection rule sets for cloud-based infrastructure assets, onboarding these into IDP Security Incident Event Monitoring platform.
- Established forensics capabilities to support whistle blower investigations globally.
- Presented on the progress against the Cyber Uplift Program to the IDP Board and led a board level incident response exercise focusing on key decisioning around recovery and response.
- Delivered NIST Maturity Uplift across the following domains (maturity uplift results determined by external vendor audit)
 - Protect 1.90 (2022) – 2.7 (2025)
 - Detect 1.30 (2022) – 3.2 (2025)
 - Respond 2.08 (2022) – 3.6 (2025)

PwC

Senior Manager Defensive Security July 2022

Dimensions and Key Achievements

- Security Operations secondment to a multinational mining company for a five-month period reporting directly to the Global Vice President, with responsibilities that include
 - Identifying and delivering on opportunities to uplift the company's Cyber Threat Intelligence function, Security Operations – Security Orchestration Automation Response (SOAR) tooling implementation, detection, and response capability.
- Development of operating models for incident response capabilities, where the capability is responsible for servicing external and internal client stakeholders, including government agencies.
- Incident responder responsible for leading client engagements for organisations large and small to detect, contain, and recover from cyber security incidents. Responding to several major ransomware incidents, as well as incidents involving business email compromise and accidental data exposure.
- Development of thought leadership pieces with respect to security operations, ransomware in the supply chain.
- Delivery of presentations to board members on contemporary cyber threats organisations are facing, bringing real world experience to the conversation to improve organisations preparedness to response to cyber incidents.
- Member of the 'She's Here' in Technology working group and was responsible for the coordination and delivery of PwC International Women's Day campaign for 2023. In addition to this I mentor female STEM graduates across PwC globally supporting females in reaching their full potential in the workforce.

PwC

Manager Defensive Security June 2021

Dimensions and Key Achievements

- Led engagements reviewing Cybersecurity submissions to regulatory bodies to validate funding requirements to uplift Cyber maturity in line with regulatory frameworks.
- Delivery of post incident reviews for major ransomware incidents, supporting clients in understanding the value of reflection and helping clients to identify and capitalise on opportunities to improve capability and build resilience after a cyber incident.
- Led the development and delivery of cyber crisis simulations for top executives and boards at several ASX10 and other major organisations.
- Coaching and mentoring junior team members on development of client engagements, this included 4 months in PwC Skilled Services Hubs developing our team's defensive security capability through the development and delivery of internal training.

AGL Energy

Cybersecurity Lead – Forensics & Information Intelligence January 2020

Dimensions and Key Achievements

- Leading a team of forensic investigators and cyber security engineers.
- Delivery of eDiscovery infrastructure (Nuix) on AGL Azure platform, reducing reliance on external vendors and reduction in data leaving the AGL environment
- Leading the delivery of eDiscovery RFP identifying best in breed external suppliers to work on AGL Nuix infrastructure to continue to develop eDiscovery capability at AGL
- Implementation of Azure Information Protection for all AGL users
- Responsible for both CASB DLP, Zscaler DLP infrastructure and ongoing policy development, refinement
- Developing information intelligence capability at AGL utilising industry leading technologies, OSINT and threat hunting
- Lead forensics investigations (including fraud, code of conduct violations) in coordination with internal teams, associated 3rd parties and government agencies.
- Collection and analysis of digital evidence in case of regulator investigation, legal request, or insider investigation
- Identify and analyse key pieces of intelligence information to provide input to business led security investigations
- Conduct forensics investigations into cyber related incidents within AGL.

AGL Energy

Cybersecurity Lead 3rd Party Security Management October 2018

Dimensions and Key Achievements

- Development of AGL strategy for management of existing and new 3rd party vendors
- Assessing and reporting on the security posture of AGL security relative to data security and fraud prevention controls
- Lead internal investigations on suspected or actual internal fraud activities this includes coordinating examining of technology pertinent to the investigation, stakeholder engagement, remediation strategy and reporting close out.
- Ensure that Security Incidents relating to data loss are captured in accordance with the Security Incident Management process. Monitor for Root cause analysis and remediation actions and assigned to owners to close the Incident with evidence.
- Provide reports to AGL's Privacy Officer where the security of AGL's data is weak or presenting risk of breach, work with the Privacy Officer and the business on treatment plans.

Relevance to GDAC-SA Tender

Annemarie brings over 10 years of cybersecurity leadership experience with proven capability in establishing Security Operations Centres, implementing NIST Cybersecurity Framework, and leading incident response for multinational organisations. Her expertise in securing cloud-based infrastructure, developing detection and response capabilities, and coordinating global security operations is directly applicable to protecting the GDAC-SA Advanced Analytics Platform and Saudi Arabia's sensitive geological datasets.

Key qualifications include:

- Established Security Operations Centre including incident response plans, playbooks, and analyst training
- Led establishment of Cyber Delivery Centre with 18 experienced cybersecurity practitioners
- Delivered measurable NIST maturity uplift: Protect (1.90→2.7), Detect (1.30→3.2), Respond (2.08→3.6)
- Developed detection rule sets for cloud-based infrastructure assets and SIEM integration
- Led global rollout of Endpoint Detection and Response toolset for servers and workstations
- Established Global Crisis Management Framework working with senior executives
- Led incident response for major ransomware incidents, business email compromise, and data breaches
- Security Operations secondment to multinational mining company implementing SOAR tooling and threat intelligence
- Delivered eDiscovery infrastructure (Nuix) on Azure platform with forensics capabilities
- Board-level presentations on cyber threats and incident response exercises
- Master of IT in Cybersecurity from Charles Sturt University

Nilesh Khadse

GIS Full Stack Developer



Experience

Senior Full Stack Engineer (GIS)

Deimos Space UK · Full-time

Jun 2021 - Present · 4 yrs 7 mos

Harwell Campus, England, United Kingdom

- Developed numerous Earth Observation applications, involving the creation of APIs, user interfaces, and processing large volumes of satellite data.
- Collaborated with the European Space Agency to successfully deliver several projects, including:
 - Designed a space debris collision risk assessment web application for satellite operators.
 - Implemented an Authentication and Access Management (IAM) service using Keycloak OpenID and an Accounting system for a microservices platform catering to Earth Observation application providers. This microservice plays a central role in the EU's Destination Earth (DestinE) initiative.

Skills: Python, docker and +4 skills

Senior MetOcean Software Developer

Fugro · Full-time

Nov 2019 - May 2021 · 1 yr 7 mos

Oxfordshire, England, United Kingdom

- Successfully implemented the Apache Airflow ETL platform for data processing workflows.
- Developed an atmospheric lightning data harvester in Python using the asyncio library, achieving a significant reduction in the server's CPU usage from 25% to 0.1% compared to its predecessor.
- Transformed the data ingestor application from a monolith Windows service to a containerized .NET Core application. This initiative not only reduced reliance on an aging data ingestion system but also made the team platform agnostic.
- Set up Elastic (ELK) analytics platform for monitoring workflows and applications, contributing to improved visibility and performance tracking.

Skills: Geoprocessing, Elastic Stack (ELK) and +3 skills

Principal Software Developer

Neftex Exploration Insights · Full-time

Jun 2014 - Oct 2019 · 5 yrs 5 mos

Oxford, United Kingdom

- Ideated and developed the 'QuickPlates' web mapping application, showcasing reconstructed data through geological times. This application also provided a client data processing service, marking a notable achievement.
- Devised a geological data reconstruction algorithm based on scientific literature, enabling the provision of 'reconstruction as a service' for client data.
- Developed a tool utilizing the OpenCV library to extract geographical contour features from historical map images.
- Built an aesthetic geological timescale using D3.js.
- Developed numerous ArcGIS plugins, web mapping applications and geospatial ETL workflows using FME.

Skills: React.js, .NET Framework and +3 skills

Senior GIS Analyst

HR Wallingford (SeaZone Solutions)

Apr 2013 - Jul 2014 · 1 yr 4 mos

SeaZone Solutions

4 yrs 7 mos

GIS Analyst

Jan 2009 - Mar 2013 · 4 yrs 3 mos

GIS Technician

Sep 2008 - Dec 2008 · 4 mos

Education

The University of Edinburgh

MSc, GIS

2007 - 2008

K. K. Wagh Institute of Engineering Education

B. Tech, Agricultural Engineering

2003 - 2007

Relevance to GDAC-SA Tender

Nilesh brings 16+ years of GIS and geospatial software development experience with proven expertise in building Earth Observation applications, web mapping platforms, and large-scale spatial data processing systems. His work with the European Space Agency on the DestinE initiative and development of geological reconstruction applications demonstrates capability directly applicable to the GDAC-SA Advanced Analytics Platform.

Key qualifications include:

- MSc in GIS from The University of Edinburgh
- Senior Full Stack Engineer (GIS) at Deimos Space UK collaborating with European Space Agency
- Developed Earth Observation applications involving APIs, user interfaces, and processing large volumes of satellite data
- Designed space debris collision risk assessment web application for satellite operators
- Implemented Authentication and Access Management (IAM) service using Keycloak OpenID for EU's Destination Earth (DestinE) initiative
- Built QuickPlates web mapping application for reconstructed geological data through geological time periods
- Devised geological data reconstruction algorithm based on scientific literature
- Developed numerous ArcGIS plugins, web mapping applications and geospatial ETL workflows using FME
- Implemented Apache Airflow ETL platform for data processing workflows
- Achieved 99.6% reduction in server CPU usage through optimized Python asyncio implementation
- Experience with Python, Docker, React.js, .NET Framework, ArcGIS, FME, D3.js, OpenCV, Elastic Stack (ELK)
- 16+ years combined experience in GIS analysis, software development, and geospatial systems (2008-present)